Backend Schema & Database Architecture

1. Database Strategy

The platform utilizes a polyglot persistence strategy to handle different data access patterns efficiently:

- **PostgreSQL:** Primary relational database for Users, Roles, Assessments, and Interview Sessions.
- **MongoDB:** NoSQL document store for high-volume, unstructured telemetry data, CV tracking logs, and AI transcriptions.
- **Redis:** In-memory data store for real-time WebSocket session management and temporary state caching.
- **Amazon S3:** Object storage for video/audio recordings and ID verification images.

2. Entity-Relationship Diagram (ERD)

```
erDiagram
    USERS ||--o{ ASSESSMENTS : creates
    USERS {
        uuid id PK
        string email
        string password_hash
        string role "RECRUITER, ADMIN, CANDIDATE"
        string preferred_language
        string timezone
        datetime created_at
    }

    CANDIDATES ||--o{ SESSIONS : participates
    CANDIDATES {
        uuid id PK
        string first_name
        string last_name
        string email
        string native_language
        string region_code "e.g., EU, US, AP"
        string resume_url
        boolean accommodations_requested
    }

    ASSESSMENTS ||--o{ SESSIONS : configures
    ASSESSMENTS {
        uuid id PK
        uuid recruiter_id FK
        string title
        jsonb questions
        int time_limit_minutes
        string security_level "STANDARD, STRICT"
    }

    SESSIONS ||--|| SESSION_RESULTS : generates
    SESSIONS {
```

```
        uuid id PK
        uuid assessment_id FK
        uuid candidate_id FK
        string data_residency_zone "e.g., eu-central-1"
        datetime start_time
        datetime end_time
        string status "PENDING, IN_PROGRESS, COMPLETED"
        string video_s3_url
    }

    SESSION_RESULTS {
        uuid id PK
        uuid session_id FK
        int ai_technical_score
        int cheating_probability_score
        jsonb transcript_analysis
    }
```

3. Telemetry & Biometrics Schema (MongoDB)

Since tracking data is generated continuously (e.g., every frame or keystroke), it is stored in MongoDB as time-series documents.

Collection: `tracking_logs`

```
{
  "_id": "ObjectId",
  "session_id": "uuid",
  "timestamp": "ISODate",
  "event_type": "GAZE_DEVIATION | HAND_OUT_OF_FRAME | AUDIO_ANOMALY | KEYSTROKE_ANOMALY",
  "severity": "LOW | MEDIUM | HIGH",
  "details": {
    "gaze_angle": 32.5,
    "duration_ms": 1500,
    "snapshot_s3_url": "s3://bucket/snapshots/uuid.jpg"
  }
}
```

Collection: `cognitive_fingerprints`

```
{
  "_id": "ObjectId",
  "candidate_id": "uuid",
  "session_id": "uuid",
  "keystroke_metrics": {
    "average_dwell_time_ms": 85,
    "average_flight_time_ms": 120,
    "variance": 15.2
  },
  "mouse_metrics": {
    "movement_linearity_score": 0.85
```

```
  },
  "is_bot_flag": false
}
```

4. Data Security & Encryption Strategy

1. **PII and Biometric Data Encryption:** All Personally Identifiable Information (PII) in PostgreSQL is encrypted at rest using AES-256. Biometric baseline data (facial scans, voice prints) are irreversibly hashed and never stored in plain text.
2. **Ephemeral Telemetry:** Detailed tracking logs (MongoDB) are retained only for the duration of the hiring cycle. An automated cron job purges telemetry data 30 days after the `SESSION_RESULTS` are marked as reviewed, leaving only the aggregated `cheating_probability_score`.
3. **Signed URLs:** Access to S3 video recordings and snapshots is strictly controlled via short-lived pre-signed URLs generated by the API Gateway, ensuring that media assets cannot be accessed publicly.